

Problem

A three-phase load consists of three 100- Ω resistors that can be wye- or delta-connected. Determine which connection will absorb the most average power from a three-phase source with a line voltage of 110 V. Assume zero line impedance.

Step-by-step solution

Step 1 of 2

For the WYE – connected load,

$$I_p = I_L, \quad V_L = \sqrt{3} V_p, \quad I_p = \frac{V_p}{Z}$$

$$S = 3 V_p I_p^* = \frac{3 |V_p|^2}{Z^*} = \frac{V_L |\sqrt{3}|^2}{Z^*}$$

$$S = \frac{|V_L|^2}{Z^*} = \frac{(110)^2}{100} = 121 \text{ W}.$$

Step 2 of 2

For the delta connected load,

$$V_p = V_L, \quad I_L = \sqrt{3} I_p, \quad I_p = \frac{V_p}{Z}$$

$$S = 3 V_p I_p^* = \frac{3 |V_p|^2}{Z^*} = \frac{3 |V_L|^2}{Z^*}$$

$$S = \frac{(3)(110)^2}{100}$$

$$= 363 \text{ W}$$

This shows that the delta connected load will deliver three times more average power than the WYE – connected load.

This is also evident from,

$$Z_y = \frac{Z_\Delta}{3}.$$